

Reflective Journal

L02 Exploring Deep Learning Tools: A No-Code Introduction to TensorFlow and Keras

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The goal of not having to write code was missed since I had to write a little code to get it to work. I needed to create a sample image on my Google Drive and then direct it there. Once that was corrected, then the lab ran as intended.

I did see how the model ran and made predictions though I didn't train it myself. Since there are images from imagenet, when I loaded my own image, it was able to recognize patterns and produced class probabilities. It didn't really learn anything new because it just applied patterns that it already knew.

Data preprocessing was instrumental. VGG16 expects images to be formatted in a very specific way – resized to 224x224 which it had to do to my own image, converted to arrays, expanded into a batch dimension, and normalized using the model's preprocessing function. Without these steps the model will throw errors, or produce meaningless predictions/nonsense. Preprocessing is basically the translation layer that makes the input understandable for the model.

The model is sensitive to input modifications. Brightness, noise, cropping or angle can shift the predictions. The model will do best predicting images similar to what it was trained on. When the input is different, the confidence drops or changes the predicted class completely.

Overall, the lab showed me that pre-trained models work but that they rely upon proper preprocessing and stable input quality – and location of the sample image properly identified! GIGO – model predictions are only as good as the data that are fed into the pipeline.